Course Code: CSE3703	**Course Name: Cryptography**	
Credits: 3 (3-0-0)	**Contact Hours: 3** hours Theory per week	
Batch: 2023, 6th Sem **Academic Year:** 2025-2026	**Semester Duration:** 19th Jan,2026 to 8th May 2026	
Course Faculty: Dr. Satyendr Singh [8683856238*: satyendr.singh@bmu.edu.in*] Dr. Monika Mokan [9694934358*: monika.mokan@bmu.edu.in*]	**Course Coordinator:** **Name:** Dr. Satyendr Singh **Email:***satyendr.singh@bmu.edu.in* **Office:** IV Floor, E2 Building	

Aim of the course: The goal of this course is to understand the principles and practice of cryptography algorithms and fundamentals of security protocols and techniques. It covers cryptography, network-based security threats and vulnerabilities, and practical solutions to system and security.

Course Overview and Context: Cryptography is a VI semester core course in computer science and engineering undergraduate program. This course focuses on the principles and applications of cryptographic techniques. This course will provide overview of classical encryption techniques, block ciphers and the data encryption standard, finite fields, advanced encryption standard, confidentiality using symmetric encryption, public-key cryptography, key management, hash and MAC algorithms, digital signatures, and authentication applications.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand the fundamental concepts in number theory, field theory, and cryptography.	Employability	→ Intensive theory sessions on number theory, modular arithmetic, and cryptographic foundations. → In-class explanations of mathematical tools used in cryptography. → Concept-based assignments and close-notebook examinations.
CO2	Analyze the principles of symmetric and asymmetric encryption algorithms.	Skill Development	→ In-depth classroom discussions on symmetric and asymmetric encryption algorithms. → Algorithm walkthroughs and step-by-step encryption/decryption demonstrations. → Quiz-based assessment using MCQs to test analytical understanding.
CO3	Apply cryptographic techniques to solve practical problems related to data confidentiality, integrity, and authentication.sensors,	Skill Development	→ Problem-solving sessions based on encryption, hashing, and digital signature techniques. → Assignment-based application of cryptographic methods to real-world security scenarios.

	actuators and cloud service platforms.		
CO4	Evaluate different cryptographic algorithms based on their strengths, weaknesses, and suitability for various applications.	Entrepreneurship, Skill Development	→ Comparative analysis of cryptographic algorithms for secure communication systems. → Evaluation-focused questions in end-semester examination.

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Number Theory and Finite Fields Modular arithmetic, Euclidian Algorithm, Primality Testing, Fermat's and Euler's theorem, Chinese Reminder theorem, Discrete Logarithms, Finite Fields.	7	CO1
Basic of Cryptography Cryptography and basic terminologies, Security goals, Security Thr and Vulnerabilities, Security Services, Security Mechanisms	3	CO2
Classical Encryption Techniques, Symmetric Encryption- Basic Theory of Enciphering, Stream Cipher, Block Encryption Algorithms: DES, AES, Block Cipher Modes of operation.	7	CO2
Asymmetric key cryptographic techniques: principles – RSA – ElGamal - Elliptic Curve cryptography – Key distribution and Key exchange protocols	7	CO2
Message Authentication Code (MAC), Hash functions, Secure Hash Algorithm: SHA-1, MD5, Keyed-Hash Message Authentication Code (HMAC).	7	CO3
Digital Signature Digital Signature Standard, Digital Signature Algorithm, RSA Digital Signature, Elliptic Curve Digital Signature algorithm.	6	CO3
User Authentication Mechanisms Authentication Fundamentals, Passwords, Authentication Tokens, Certificate-Based Authentication, Biometric Authentication, Kerberos	5	CO3
Historical Cryptanalysis, Cryptanalysis of block ciphers, Brute force attack, Standard ASCII attack, Man in the middle attack, Cryptanalysis of hash functions.	6	CO4

CO/PO Mapping:

CO/PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3					2		1					3			
CO2		1					3				2		1	2		
CO3		3	2				3				1		1	2	3	
CO4		3	2									2	2			

Learning Resources:

Textbook: William Stallings: Cryptography and Network Security, 7th Edition, Prentice-Hall / Pearson Education, Englewood Cliffs /New Delhi, 2017.

Reference Books:

1. Michael Goodrich & Roberto Tamassia, ***"Introduction to Computer Security"***, Prentice-Hall/ Pearson Education, Englewood Cliffs/ New Delhi.
2. Behrouz A Forouzan, "***Cryptography and Network security***", 3rd Edition, McGraw Hill Education.
3. Atul Kahate, ***"Cryptography and Network Security",*** Tata McGraw Hill Education.

Internet Links (annotated): Following are the online available resources:

MIT equivalent course (Network and Computer Security): https://ocw.mit.edu/courses/electrical- engineering-and-computer-science/6-857-network-and-computer-security-spring-2014/index.htm

NPTEL equivalent course (Cryptography and Network Security): http://nptel.ac.in/courses/106105031/

Experiential Learning Component (20 %):

Assignments will include challenging cryptography puzzles and riddles, fostering critical thinking and problem-solving abilities.

Assessment Pattern:

The final grade will be based on the marks / grades obtained in the mid-semester and end-semester evaluation along with other assessments defined below. Relative grading method defined in the academic regulations of the university will be followed to grade the students. Student has to secure minimum 40% of marks after completing all the assessments in the following table to become eligible for grading.

Evaluation Component	**Weightage (%)**	**Evaluation Schedule/Timeline**	**Rubrics/Remarks**	**CO mapping**
Assignment	20%	March 1st week	Conceptual clarity, correctness of cryptographic methods, quality of explanation, originality, and timely submission will be evaluated.	CO1, CO2
Mid Semester Examination	20%	As per academic calendar	Evaluation will focus on conceptual understanding, analytical ability, and problem-solving skills based on covered syllabus.	CO1, CO2, CO3
Quiz	20%	April 1st week	MCQ-based assessment to test understanding of key concepts, algorithms, and terminologies discussed in class.	CO2, CO3

End Term Evaluation	40%	As per academic calendar/Date-sheet	Comprehensive assessment of theoretical knowledge, analytical depth, and ability to evaluate cryptographic techniques across the entire syllabus.	CO1, CO2, CO3, CO4

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.